Honours Project Midterm Report

Project Title: Evaluating Adversarial Robustness in Medical Image Classification

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1. Aims, Objectives, and SMART Goals

Deep learning has demonstrated remarkable success in medical image classification, particularly in tasks such as chest X-ray disease detection. Despite their strong predictive performance, these models are known to be highly vulnerable to adversarial perturbations—small, often imperceptible changes to the input that can lead to incorrect predictions. Such vulnerabilities present significant concerns for the clinical reliability and safety of AI-assisted diagnostic systems. The principal aim of this project is to evaluate the adversarial robustness of deep learning–based chest X-ray classifiers, analysing how different types and magnitudes of adversarial attacks influence model accuracy and decision behaviour. The ultimate goal is to provide quantitative and visual evidence of robustness limitations, thereby contributing to the broader understanding of trustworthy and dependable medical AI.

To achieve this aim, a sequence of specific objectives has been established:

1. Conduct a comprehensive literature review covering medical image classification methods, convolutional neural network architectures, and established adversarial attack algorithms such as FGSM, PGD, and C&W.
2. Implement an initial prototype experiment using a smaller and simpler chest X-ray dataset to validate core components of the experimental pipeline, including data preprocessing, model training, evaluation, and basic adversarial example generation.
3. Prepare the larger and clinically representative NIH ChestX-ray14 dataset by performing label parsing, binary task construction, data cleaning, preprocessing and train/validation/test splitting.
4. Train a baseline convolutional neural network—such as ResNet18 or CheXNet—on the NIH dataset and obtain clean test-set accuracy to serve as a benchmark for robustness evaluation.
5. Implement one or more standard adversarial attack methods and validate their behaviour on the trained baseline model across varying perturbation strengths.
6. Perform a systematic robustness evaluation by generating accuracy–distortion curves and comparing the degradation patterns resulting from different attack methods.
7. Produce visualisations of adversarial examples and analyse failure modes, discussing their implications for real-world diagnostic safety and the deployment of AI in clinical workflows.

These objectives have been translated into SMART goals:

- Literature Review: Produce a synthesised review of medical image classification models and adversarial attack techniques, including FGSM, PGD and C&W, by Week 5.
- Prototype Experimentation: Implement a complete miniature pipeline using a small chest X-ray dataset, validating data processing, model training and basic adversarial example generation by Week 10.

- Dataset Preparation: Complete the full preprocessing of the NIH ChestXray14 dataset—including label parsing, binary construction and dataset partitioning—and verify successful data loading by Week 13.
- Baseline Model Training: Train a pre-trained ResNet18 or CheXNet classifier on the NIH dataset and obtain validated clean accuracy results by Week 15.
- Attack Implementation: Implement FGSM, PGD and at least one optimisation-based attack (e.g.,C&W), ensuring stable generation of adversarial samples across multiple perturbation levels by Week 17.
- Robustness Evaluation: Produce full accuracy–epsilon or accuracy–L_2 curves, along with quantitative robustness tables comparing attack strength and model performance, by Week 20.
- Analysis: Generate adversarial visualisations, identify vulnerability patterns, and complete the first full draft of the results and analysis section by Week 23, ahead of project completion.

2. Summary of Work Completed

The work completed so far represents steady, well-structured, and technically substantive progress, establishing a strong foundation for the subsequent stages of the project. The early phase of the work focused on developing a solid theoretical understanding of adversarial vulnerabilities in medical image classification systems. Through a comprehensive review of recent literature, I examined how deep learning models deployed in clinical imaging tasks can be compromised by carefully crafted adversarial perturbations, even when such perturbations are visually imperceptible to human observers. Particular attention was given to convolutional neural network architectures that are widely adopted in medical imaging research, including ResNet and DenseNet, with an emphasis on their structural characteristics, representational capacity, and reported performance in chest X-ray classification tasks. Studies analysing FGSM, PGD, and C&W attacks provided concrete evidence that standard accuracy metrics alone are insufficient to characterise model reliability in safety-critical environments, reinforcing the need for systematic and quantitative robustness evaluation.

In parallel with the literature review, significant effort was devoted to establishing a robust and reproducible experimental environment. This involved configuring the PyTorch and CUDA software stack, resolving dependency and version compatibility issues, and ensuring stable GPU execution. The overall project directory structure was carefully organised to separate datasets, trained models, experimental scripts, and evaluation outputs, thereby supporting scalable experimentation and transparent result management. These engineering-level preparations were not only essential for ensuring experimental stability, but also played an important role in enabling reproducibility and facilitating subsequent large-scale evaluation and analysis.

The methodological development of the project followed a deliberate, staged approach. To validate the correctness of the experimental pipeline, I first conducted a prototype implementation using a small and simplified chest X-ray dataset. This stage served as a controlled environment for verifying all core components of the workflow. In particular, I systematically validated data preprocessing steps, including resizing all images to a fixed resolution of 224×224 pixels, normalising pixel intensities to standard ranges, and confirming the correctness of data loading, batching, and shuffling procedures within PyTorch. A preliminary ResNet model was trained to assess training stability, convergence behaviour, and baseline classification performance. Following this, the FGSM attack was implemented

and applied to the trained model to generate basic adversarial examples. These initial experiments were critical for identifying implementation issues, validating gradient computation and loss backpropagation, and refining experimental design choices before transitioning to more complex settings.

Building on the validated prototype, the experimental framework was progressively extended to support systematic and fair robustness evaluation. A baseline ResNet18 model was first trained on the simplified dataset to establish reference performance and to verify evaluation metrics such as classification accuracy and confusion matrices. The framework was then expanded to include DenseNet121, allowing a direct comparison between architectures under identical experimental conditions. Evaluation on clean data demonstrated that DenseNet121 achieved higher classification performance, motivating a deeper investigation into whether improved clean accuracy translates to improved robustness under adversarial perturbations.

To enable meaningful and fair comparisons across attack methods and model architectures, unified L2-constrained variants of FGSM, PGD, and C&W attacks were implemented. Particular care was taken to explicitly control the perturbation norm in each attack, ensuring that all adversarial examples satisfied the same L2 constraint. This design choice was motivated by the need to avoid misleading conclusions that may arise from inconsistent perturbation budgets. Using this unified framework, accuracy–distortion curves were generated for both ResNet18 and DenseNet121, with distortion measured by the L2 norm of the applied perturbation. The clean baseline point corresponding to L2 = 0 was explicitly included in the curves, allowing a continuous and interpretable view of model performance degradation as perturbation strength increased. The resulting curves confirmed the correctness of the attack implementations and revealed clear and systematic differences in robustness behaviour between the two architectures.

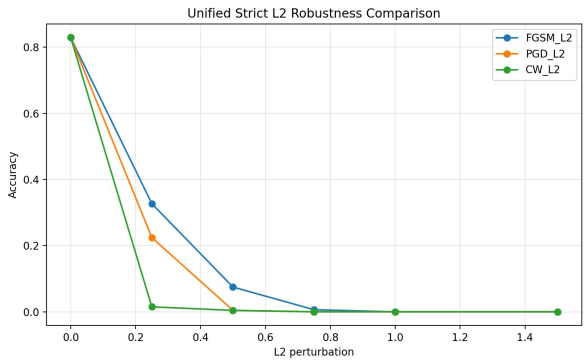

Figure 1. Unified L2 robustness curves for the ResNet18 model under FGSM, PGD and C&W attacks.The figure is generated by the author based on experimental results.

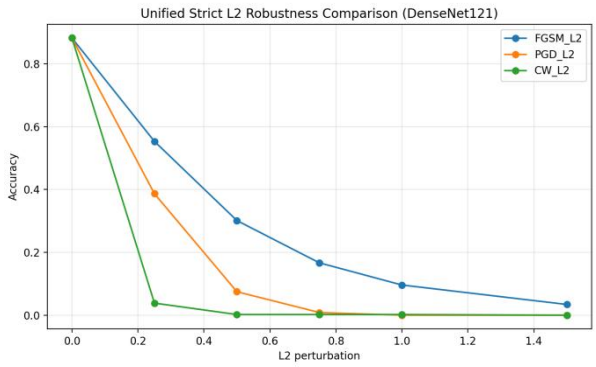

Figure 2. Unified L2 robustness curves for the DenseNet121 model under FGSM, PGD and C&W attacks.The figure is generated by the author based on experimental results.

In addition to attack implementation and evaluation, substantial progress was made in preparing the primary dataset for the main phase of the project. The full NIH ChestX-ray14 dataset was downloaded and organised, comprising over 100,000 chest X-ray images distributed across twelve directories. The accompanying metadata file was parsed to construct a binary classification task distinguishing between “No Finding” and “Disease” cases. Data cleaning procedures were applied to remove incomplete or ambiguous entries, and the dataset was split into training, validation, and test sets following established practices in chest X-ray research. Custom PyTorch Dataset and DataLoader components were implemented to support

efficient large-scale image loading, on-the-fly preprocessing, and batch construction. These components were rigorously tested to confirm their correctness and scalability.

To further verify compatibility between the experimental framework and the NIH dataset, initial baseline experiments were conducted on a subset of the data. These experiments confirmed that the preprocessing pipeline remained consistent with earlier prototype settings and that model training and inference behaved stably under the increased data complexity. The results indicated that the framework is well-prepared for full-scale robustness evaluation on the complete NIH ChestX-ray14 dataset, forming a solid basis for the subsequent stages of adversarial analysis and potential defence exploration.

3. Plan for the Remainder of the Project, including Personal Reflection and Self-Assessment

The project is broadly on track. The literature review, prototype experimentation and full preparation of the NIH ChestX-ray14 dataset have been completed as planned. The remaining work—baseline model training, implementation of FGSM/PGD/C&W attacks, robustness evaluation and visualisation—fits within the scheduled Weeks 14–22. The dissertation writing period (Weeks 23–26) and contingency buffer (Weeks 27–28) provide sufficient time to complete the project without delays.

The main risks concern computational demands and the stability of adversarial attack implementations. Training on the NIH dataset and running iterative attacks such as PGD or C&W may be time-consuming. This will be mitigated through staged experiments, reduced batch sizes when needed and verifying attack behaviour on small subsets before scaling up. Unexpected model behaviour or limited robustness differences may also arise; additional metrics and visual analysis (e.g., Grad-CAM) will be used if needed to strengthen the findings.

The completed work has strengthened my understanding of CNN-based medical image analysis and adversarial robustness. Building the prototype pipeline before working with the full NIH dataset proved invaluable for debugging and improving the workflow. I have gained experience in dataset management, modular design and experiment planning. If restarting the project, I would structure the pipeline even more modularly from the beginning to simplify later experimentation.

Given the progress so far and the clarity of the remaining tasks, I believe I am currently on track for an Upper Second (2:1) and potentially a First Class outcome. Achieving the higher grade will depend on producing strong robustness results and delivering a well-structured final report.